

Confidence Intervals & Hypothesis Testing

ANSWER KEY



Constructing a confidence interval

- 1 A random sample of 50 students has a mean study time of $\bar{x} = 6.2$ hours per week, with sample standard deviation $s = 1.8$ hours. Construct a 95% confidence interval for the population mean study time, using a critical value of $z^* \approx 1.96$.

Standard error: $\frac{s}{\sqrt{n}} = \frac{1.8}{\sqrt{50}} \approx 0.255$. Margin of error: $1.96(0.255) \approx 0.499$. Confidence interval: 6.2 ± 0.499 , or approximately (5.70, 6.70) hours.

- 2 Interpret the confidence interval from the previous problem in the context of the study time question, using correct statistical language.

We are 95% confident that the true mean weekly study time for the population of all students is between approximately 5.70 and 6.70 hours — meaning that if we repeated this sampling process many times, about 95% of the resulting confidence intervals would capture the true population mean.

Margin of error and sample size

- 3 Explain two ways to decrease the margin of error of a confidence interval, and state a tradeoff involved with each.

(1) Increase the sample size n : since margin of error is proportional to $\frac{1}{\sqrt{n}}$, a larger sample shrinks the margin of error, but larger samples cost more time and resources to collect. (2) Decrease the confidence level (e.g. from 95% to 90%): a lower confidence level uses a smaller critical value, shrinking the margin of error, but this means being less confident that the interval actually captures the true parameter.

The logic of hypothesis testing

- 4 State the null hypothesis H_0 and alternative hypothesis H_a for a test of whether a coin is biased toward heads (true proportion of heads p), and classify the test as one-tailed or two-tailed.

$H_0 : p = 0.5$ (the coin is fair). $H_a : p > 0.5$ (the coin is biased toward heads). This is a one-tailed (right-tailed) test, since the alternative specifies a direction.

- 5 Explain what a p -value represents, in the context of a hypothesis test.

The p -value is the probability of obtaining a test statistic at least as extreme as the one observed, assuming the null hypothesis is true. A small p -value suggests the observed data would be unusual if H_0 were true, providing evidence against H_0 .

Four-step hypothesis testing procedure

- 6 A manufacturer claims their light bulbs last on average $\mu = 1000$ hours. A sample of 40 bulbs has mean lifetime $\bar{x} = 985$ hours with sample standard deviation $s = 45$ hours. State the hypotheses and compute the test statistic for testing whether the true mean lifetime is less than claimed.

$$H_0 : \mu = 1000. H_a : \mu < 1000. \text{ Test statistic: } z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{985 - 1000}{45/\sqrt{40}} = \frac{-15}{7.115} \approx -2.11.$$

- 7 Using the test statistic $z \approx -2.11$ from the previous problem and a p -value of approximately 0.017, state a conclusion at the $\alpha = 0.05$ significance level.

Since the p -value (0.017) is less than $\alpha = 0.05$, we reject H_0 . There is convincing statistical evidence that the true mean bulb lifetime is less than the manufacturer's claimed 1000 hours.

Type I and Type II errors, and power

- 8 In the light bulb test above, describe what a Type I error would mean in context, and what a Type II error would mean in context.

A Type I error would mean concluding the true mean lifetime is less than 1000 hours when it actually is not (rejecting a true H_0) — wrongly penalizing the manufacturer. A Type II error would mean failing to detect that the true mean lifetime really is less than 1000 hours (failing to reject a false H_0) — missing a real problem with the bulbs.

- 9 Explain what the power of a test measures, and state one way to increase it.

Power measures the probability that a test correctly rejects H_0 when the alternative hypothesis is actually true (i.e. correctly detecting a real effect). Power can be increased by increasing the sample size, which reduces variability in the sampling distribution and makes real effects easier to detect.

Confidence intervals for a proportion

- 10 A poll of 400 randomly selected voters finds that 220 support a candidate. Construct a 95% confidence interval for the true population proportion, using $z^* \approx 1.96$.

$$\hat{p} = \frac{220}{400} = 0.55. \text{ Standard error: } \sqrt{\frac{0.55(0.45)}{400}} = \sqrt{\frac{0.2475}{400}} \approx 0.0249. \text{ Margin of error: } 1.96(0.0249) \approx 0.0488. \text{ Confidence interval: } 0.55 \pm 0.0488, \text{ or approximately } (0.501, 0.599).$$

Determining sample size for a target margin of error

- 11 A researcher wants a margin of error of at most 0.03 for a 95% confidence interval for a population proportion, using a conservative planning value of $\hat{p} = 0.5$ (which maximizes required sample size). Find the minimum sample size needed, using $n = \left(\frac{z^*}{m}\right)^2 p(1-p)$ with $z^* = 1.96$.
- $$n = \left(\frac{1.96}{0.03}\right)^2 (0.5)(0.5) = (65.33)^2(0.25) \approx 4268.4(0.25) \approx 1067.1. \text{ Round up: } n = 1068 \text{ (sample size must always be rounded up to guarantee the margin of error requirement is met).}$$

One-sample t-test for a mean

- 12 A nutritionist claims a new granola bar has a mean of 200 calories. A sample of 15 bars has mean $\bar{x} = 206$ calories with sample standard deviation $s = 9$ calories. Since the population standard deviation is unknown and n is small, a t -test is used. Compute the test statistic $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$.

$$t = \frac{206 - 200}{9/\sqrt{15}} = \frac{6}{2.324} \approx 2.58.$$

- 13 Explain what it means for a hypothesis test result to be "statistically significant," and why statistical significance does not always imply practical importance.

A result is statistically significant when the p -value is small enough (below the chosen α) to reject H_0 , meaning the observed data would be unlikely if H_0 were true. However, with a very large sample size, even a tiny, practically unimportant difference can become statistically significant — so significance alone doesn't guarantee the effect is large or meaningful in real-world terms; the actual size of the effect should also be considered.

- 14 A researcher wants to test whether a new fitness program changes the average resting heart rate of participants from the population baseline of $\mu = 72$ bpm. A sample of 36 participants after the program has mean resting heart rate $\bar{x} = 68.5$ bpm with sample standard deviation $s = 10.5$ bpm. State the hypotheses (choosing a two-sided alternative), compute the test statistic, and using a p -value of approximately 0.11, state a conclusion at $\alpha = 0.05$.

$H_0: \mu = 72$. $H_a: \mu \neq 72$. Test statistic: $t = \frac{68.5 - 72}{10.5/\sqrt{36}} = \frac{-3.5}{1.75} = -2.0$. Since the p -value (0.11) is greater than $\alpha = 0.05$, we fail to reject H_0 — there is not enough evidence that the fitness program changes the average resting heart rate.

- 15 Explain the relationship between a confidence interval and a two-sided hypothesis test at the same confidence/significance level — specifically, describe how you could use a 95% confidence interval alone to decide whether to reject $H_0: \mu = \mu_0$ at $\alpha = 0.05$, without separately computing a p -value.

A 95% confidence interval and a two-sided test at $\alpha = 0.05$ are directly linked: if the hypothesized value μ_0 falls outside the 95% confidence interval, then $H_0: \mu = \mu_0$ would be rejected at $\alpha = 0.05$; if μ_0 falls inside the confidence interval, H_0 would fail to be rejected. This means a confidence interval can be used as a quick visual/numeric check for significance without needing to separately calculate a p -value.

- 16 A city council is deciding whether to fund a new public transit line, and wants to survey residents to estimate the true proportion who would use it regularly. A preliminary small survey suggests around 35% support, but the council wants a formal survey with a margin of error no larger than 2.5% at 95% confidence. Using $n = \left(\frac{z^*}{m}\right)^2 p(1-p)$ with the preliminary estimate $p = 0.35$ and $z^* = 1.96$, find the required sample size, rounding appropriately.

$n = \left(\frac{1.96}{0.025}\right)^2 (0.35)(0.65) = (78.4)^2(0.2275) \approx 6146.6(0.2275) \approx 1398.4$. Round up: $n = 1399$ residents (sample size must be rounded up to guarantee the target margin of error is achieved).

- 17 A restaurant chain wants to test whether a new menu item's average preparation time is longer than the target of 8 minutes, since customers have complained about slow service. A random sample of 30 orders for the new item has a mean preparation time of $\bar{x} = 8.7$ minutes with sample standard deviation $s = 1.8$ minutes. State the hypotheses for this one-sided test, compute the test statistic, and using a p -value of approximately 0.019, state a conclusion at $\alpha = 0.05$ in the context of the restaurant's concern.

$H_0: \mu = 8$. $H_a: \mu > 8$. Test statistic: $t = \frac{8.7 - 8}{1.8/\sqrt{30}} = \frac{0.7}{0.329} \approx 2.13$. Since the p -value (0.019) is less than $\alpha = 0.05$, we reject H_0 . There is convincing statistical evidence that the true mean preparation time for the new menu item exceeds the 8-minute target, supporting the customers' complaints about slow service.

Multiple choice — easy 15

- 18 A sample of $n = 121$ has mean $\bar{x} = 393$. A claimed population mean is $\mu_0 = 409$, with sample standard deviation $s = 62$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.

- a. 2.84
- b. -2.34
- c. -3.34
- d. -2.84 ✓

$s/\sqrt{n} = 62/11 = 5.64$. $z = (393 - 409)/5.64 = (-16)/5.64 \approx -2.84$.

- 19 A sample of $n = 49$ has mean $\bar{x} = 264$. A claimed population mean is $\mu_0 = 265$, with sample standard deviation $s = 75$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.

a. -0.09 ✓

b. 0.41

c. 0.09

d. -0.59

$$s/\sqrt{n} = 75/7 = 10.71. z = (264 - 265)/10.71 = (-1)/10.71 \approx -0.09.$$

- 20 A sample of $n = 25$ has mean $\bar{x} = 513$. A claimed population mean is $\mu_0 = 495$, with sample standard deviation $s = 75$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.

a. -1.20

b. 0.70

c. 1.70

d. 1.20 ✓

$$s/\sqrt{n} = 75/5 = 15.00. z = (513 - 495)/15.00 = (18)/15.00 \approx 1.20.$$

- 21 A sample of $n = 121$ has mean $\bar{x} = 371$. A claimed population mean is $\mu_0 = 370$, with sample standard deviation $s = 13$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.

a. 1.35

b. 0.35

c. -0.85

d. 0.85 ✓

$$s/\sqrt{n} = 13/11 = 1.18. z = (371 - 370)/1.18 = (1)/1.18 \approx 0.85.$$

- 22 A sample of $n = 25$ has mean $\bar{x} = 469$. A claimed population mean is $\mu_0 = 471$, with sample standard deviation $s = 51$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.

a. -0.20 ✓

b. 0.30

c. -0.70

d. 0.20

$$s/\sqrt{n} = 51/5 = 10.20. z = (469 - 471)/10.20 = (-2)/10.20 \approx -0.20.$$

- 23 A sample of $n = 121$ has mean $\bar{x} = 453$. A claimed population mean is $\mu_0 = 437$, with sample standard deviation $s = 40$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.

a. 3.90

b. 4.90

c. 4.40 ✓

d. -4.40

$$s/\sqrt{n} = 40/11 = 3.64. z = (453 - 437)/3.64 = (16)/3.64 \approx 4.40.$$

- 24 A sample of $n = 16$ has mean $\bar{x} = 157$. A claimed population mean is $\mu_0 = 183$, with sample standard deviation $s = 34$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.

a. -2.56

b. -3.06 ✓

c. 3.06

d. -3.56

$$s/\sqrt{n} = 34/4 = 8.50. z = (157 - 183)/8.50 = (-26)/8.50 \approx -3.06.$$

- 25 A sample of $n = 49$ has mean $\bar{x} = 408$. A claimed population mean is $\mu_0 = 406$, with sample standard deviation $s = 15$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.

- a. 0.43
- b. -0.93
- c. 0.93 ✓
- d. 1.43

$$s/\sqrt{n} = 15/7 = 2.14. z = (408 - 406)/2.14 = (2)/2.14 \approx 0.93.$$

- 26 A sample of $n = 64$ has mean $\bar{x} = 53$. A claimed population mean is $\mu_0 = 77$, with sample standard deviation $s = 51$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.

- a. -3.76 ✓
- b. -3.26
- c. -4.26
- d. 3.76

$$s/\sqrt{n} = 51/8 = 6.38. z = (53 - 77)/6.38 = (-24)/6.38 \approx -3.76.$$

- 27 A sample of $n = 49$ has mean $\bar{x} = 187$. A claimed population mean is $\mu_0 = 162$, with sample standard deviation $s = 65$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.

- a. 2.69 ✓
- b. 3.19
- c. -2.69
- d. 2.19

$$s/\sqrt{n} = 65/7 = 9.29. z = (187 - 162)/9.29 = (25)/9.29 \approx 2.69.$$

- 28 A sample of $n = 81$ has mean $\bar{x} = 195$. A claimed population mean is $\mu_0 = 205$, with sample standard deviation $s = 58$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.

a. 1.55

b. -1.55 ✓

c. -2.05

d. -1.05

$$s/\sqrt{n} = 58/9 = 6.44. z = (195 - 205)/6.44 = (-10)/6.44 \approx -1.55.$$

- 29 A sample of $n = 121$ has mean $\bar{x} = 348$. A claimed population mean is $\mu_0 = 346$, with sample standard deviation $s = 28$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.

a. 1.29

b. 0.79 ✓

c. 0.29

d. -0.79

$$s/\sqrt{n} = 28/11 = 2.55. z = (348 - 346)/2.55 = (2)/2.55 \approx 0.79.$$

- 30 A sample of $n = 144$ has mean $\bar{x} = 125$. A claimed population mean is $\mu_0 = 131$, with sample standard deviation $s = 78$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.

a. 0.92

b. -1.42

c. -0.92 ✓

d. -0.42

$$s/\sqrt{n} = 78/12 = 6.50. z = (125 - 131)/6.50 = (-6)/6.50 \approx -0.92.$$

- 31 A sample of $n = 64$ has mean $\bar{x} = 241$. A claimed population mean is $\mu_0 = 244$, with sample standard deviation $s = 25$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.

- a. -0.46
- b. -1.46
- c. -0.96 ✓
- d. 0.96

$$s/\sqrt{n} = 25/8 = 3.12. z = (241 - 244)/3.12 = (-3)/3.12 \approx -0.96.$$

- 32 A sample of $n = 81$ has mean $\bar{x} = 448$. A claimed population mean is $\mu_0 = 459$, with sample standard deviation $s = 59$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.

- a. -2.18
- b. -1.18
- c. -1.68 ✓
- d. 1.68

$$s/\sqrt{n} = 59/9 = 6.56. z = (448 - 459)/6.56 = (-11)/6.56 \approx -1.68.$$

Multiple choice — medium 15

- 33 A random sample of $n = 121$ has mean $\bar{x} = 180$ and sample standard deviation $s = 16$. Construct a 99% confidence interval for the population mean, using $z^* \approx 2.576$.

- a. $(176.25, 181.75)$
- b. $(175.25, 184.75)$
- c. $(178.25, 183.75)$
- d. $(176.25, 183.75)$ ✓

Standard error: $s/\sqrt{n} = 16/11 = 1.45$. Margin of error: $2.576(1.45) \approx 3.75$. Interval: 180 ± 3.75 , or approximately $(176.25, 183.75)$.

- 34 A random sample of $n = 64$ has mean $\bar{x} = 93$ and sample standard deviation $s = 17$. Construct a 95% confidence interval for the population mean, using $z^* \approx 1.96$.

- a. (88.83, 97.17) ✓
- b. (87.83, 98.17)
- c. (90.83, 97.17)
- d. (88.83, 95.17)

Standard error: $s/\sqrt{n} = 17/8 = 2.12$. Margin of error: $1.96(2.12) \approx 4.17$. Interval: 93 ± 4.17 , or approximately (88.83, 97.17).

- 35 A random sample of $n = 36$ has mean $\bar{x} = 156$ and sample standard deviation $s = 23$. Construct a 90% confidence interval for the population mean, using $z^* \approx 1.645$.

- a. (149.69, 160.31)
- b. (151.69, 162.31)
- c. (148.69, 163.31)
- d. (149.69, 162.31) ✓

Standard error: $s/\sqrt{n} = 23/6 = 3.83$. Margin of error: $1.645(3.83) \approx 6.31$. Interval: 156 ± 6.31 , or approximately (149.69, 162.31).

- 36 A random sample of $n = 49$ has mean $\bar{x} = 143$ and sample standard deviation $s = 41$. Construct a 95% confidence interval for the population mean, using $z^* \approx 1.96$.

- a. (130.52, 155.48)
- b. (133.52, 154.48)
- c. (131.52, 152.48)
- d. (131.52, 154.48) ✓

Standard error: $s/\sqrt{n} = 41/7 = 5.86$. Margin of error: $1.96(5.86) \approx 11.48$. Interval: 143 ± 11.48 , or approximately (131.52, 154.48).

- 37 A random sample of $n = 144$ has mean $\bar{x} = 154$ and sample standard deviation $s = 20$. Construct a 95% confidence interval for the population mean, using $z^* \approx 1.96$.

- a. (150.73, 157.27) ✓
- b. (149.73, 158.27)
- c. (150.73, 155.27)
- d. (152.73, 157.27)

Standard error: $s/\sqrt{n} = 20/12 = 1.67$. Margin of error: $1.96(1.67) \approx 3.27$. Interval: 154 ± 3.27 , or approximately (150.73, 157.27).

- 38 A random sample of $n = 121$ has mean $\bar{x} = 40$ and sample standard deviation $s = 11$. Construct a 95% confidence interval for the population mean, using $z^* \approx 1.96$.

- a. (37.04, 42.96)
- b. (38.04, 41.96) ✓
- c. (38.04, 39.96)
- d. (40.04, 41.96)

Standard error: $s/\sqrt{n} = 11/11 = 1.00$. Margin of error: $1.96(1.00) \approx 1.96$. Interval: 40 ± 1.96 , or approximately (38.04, 41.96).

- 39 A random sample of $n = 64$ has mean $\bar{x} = 112$ and sample standard deviation $s = 35$. Construct a 95% confidence interval for the population mean, using $z^* \approx 1.96$.

- a. (103.42, 120.58) ✓
- b. (102.42, 121.58)
- c. (103.42, 118.58)
- d. (105.42, 120.58)

Standard error: $s/\sqrt{n} = 35/8 = 4.38$. Margin of error: $1.96(4.38) \approx 8.57$. Interval: 112 ± 8.57 , or approximately (103.42, 120.58).

- 40 A random sample of $n = 16$ has mean $\bar{x} = 108$ and sample standard deviation $s = 7$. Construct a 95% confidence interval for the population mean, using $z^* \approx 1.96$.

- a. (103.57, 112.43)
- b. (106.57, 111.43)
- c. (104.57, 111.43) ✓
- d. (104.57, 109.43)

Standard error: $s/\sqrt{n} = 7/4 = 1.75$. Margin of error: $1.96(1.75) \approx 3.43$. Interval: 108 ± 3.43 , or approximately (104.57, 111.43).

- 41 A random sample of $n = 25$ has mean $\bar{x} = 199$ and sample standard deviation $s = 22$. Construct a 95% confidence interval for the population mean, using $z^* \approx 1.96$.

- a. (190.38, 207.62) ✓
- b. (190.38, 205.62)
- c. (189.38, 208.62)
- d. (192.38, 207.62)

Standard error: $s/\sqrt{n} = 22/5 = 4.40$. Margin of error: $1.96(4.40) \approx 8.62$. Interval: 199 ± 8.62 , or approximately (190.38, 207.62).

- 42 A random sample of $n = 49$ has mean $\bar{x} = 53$ and sample standard deviation $s = 21$. Construct a 95% confidence interval for the population mean, using $z^* \approx 1.96$.

- a. (49.12, 58.88)
- b. (46.12, 59.88)
- c. (47.12, 58.88) ✓
- d. (47.12, 56.88)

Standard error: $s/\sqrt{n} = 21/7 = 3.00$. Margin of error: $1.96(3.00) \approx 5.88$. Interval: 53 ± 5.88 , or approximately (47.12, 58.88).

- 43 A random sample of $n = 49$ has mean $\bar{x} = 154$ and sample standard deviation $s = 59$. Construct a 90% confidence interval for the population mean, using $z^* \approx 1.645$.

a. (140.13, 167.87) ✓

b. (139.13, 168.87)

c. (140.13, 165.87)

d. (142.13, 167.87)

Standard error: $s/\sqrt{n} = 59/7 = 8.43$. Margin of error: $1.645(8.43) \approx 13.87$. Interval: 154 ± 13.87 , or approximately (140.13, 167.87).

- 44 A random sample of $n = 81$ has mean $\bar{x} = 107$ and sample standard deviation $s = 24$. Construct a 95% confidence interval for the population mean, using $z^* \approx 1.96$.

a. (101.77, 110.23)

b. (100.77, 113.23)

c. (101.77, 112.23) ✓

d. (103.77, 112.23)

Standard error: $s/\sqrt{n} = 24/9 = 2.67$. Margin of error: $1.96(2.67) \approx 5.23$. Interval: 107 ± 5.23 , or approximately (101.77, 112.23).

- 45 A random sample of $n = 36$ has mean $\bar{x} = 23$ and sample standard deviation $s = 51$. Construct a 90% confidence interval for the population mean, using $z^* \approx 1.645$.

a. (9.02, 34.98)

b. (9.02, 36.98) ✓

c. (8.02, 37.98)

d. (11.02, 36.98)

Standard error: $s/\sqrt{n} = 51/6 = 8.50$. Margin of error: $1.645(8.50) \approx 13.98$. Interval: 23 ± 13.98 , or approximately (9.02, 36.98).

- 46 A random sample of $n = 25$ has mean $\bar{x} = 125$ and sample standard deviation $s = 53$. Construct a 90% confidence interval for the population mean, using $z^* \approx 1.645$.

- a. (107.56, 142.44) ✓
- b. (107.56, 140.44)
- c. (106.56, 143.44)
- d. (109.56, 142.44)

Standard error: $s/\sqrt{n} = 53/5 = 10.60$. Margin of error: $1.645(10.60) \approx 17.44$. Interval: 125 ± 17.44 , or approximately (107.56, 142.44).

- 47 A random sample of $n = 16$ has mean $\bar{x} = 22$ and sample standard deviation $s = 56$. Construct a 95% confidence interval for the population mean, using $z^* \approx 1.96$.

- a. (-3.44, 49.44)
- b. (-5.44, 49.44) ✓
- c. (-5.44, 47.44)
- d. (-6.44, 50.44)

Standard error: $s/\sqrt{n} = 56/4 = 14.00$. Margin of error: $1.96(14.00) \approx 27.44$. Interval: 22 ± 27.44 , or approximately (-5.44, 49.44).

Multiple choice — hard 15

- 48 A poll of 400 randomly selected voters finds that 180 support a candidate. Construct a 99% confidence interval for the true population proportion, using $z^* \approx 2.576$.

- a. (0.386, 0.484)
- b. (0.416, 0.514)
- c. (0.366, 0.534)
- d. (0.386, 0.514) ✓

$\hat{p} = 180/400 = 0.450$. Standard error: $\sqrt{0.450(0.550)/400} \approx 0.025$. Margin of error: $2.576(0.025) \approx 0.064$. Interval: 0.450 ± 0.064 , or approximately (0.386, 0.514).

49 A poll of 150 randomly selected voters finds that 37 support a candidate. Construct a 99% confidence interval for the true population proportion, using $z^* \approx 2.576$.

a. (0.156, 0.307)

b. (0.136, 0.357)

c. (0.186, 0.337)

d. (0.156, 0.337) ✓

$\hat{p} = 37/150 = 0.247$. Standard error: $\sqrt{0.247(0.753)/150} \approx 0.035$. Margin of error: $2.576(0.035) \approx 0.091$. Interval: 0.247 ± 0.091 , or approximately (0.156, 0.337).

50 A poll of 150 randomly selected voters finds that 85 support a candidate. Construct a 90% confidence interval for the true population proportion, using $z^* \approx 1.645$.

a. (0.480, 0.653)

b. (0.500, 0.633) ✓

c. (0.530, 0.633)

d. (0.500, 0.603)

$\hat{p} = 85/150 = 0.567$. Standard error: $\sqrt{0.567(0.433)/150} \approx 0.040$. Margin of error: $1.645(0.040) \approx 0.067$. Interval: 0.567 ± 0.067 , or approximately (0.500, 0.633).

51 A poll of 150 randomly selected voters finds that 112 support a candidate. Construct a 90% confidence interval for the true population proportion, using $z^* \approx 1.645$.

a. (0.668, 0.825)

b. (0.688, 0.805) ✓

c. (0.688, 0.775)

d. (0.718, 0.805)

$\hat{p} = 112/150 = 0.747$. Standard error: $\sqrt{0.747(0.253)/150} \approx 0.036$. Margin of error: $1.645(0.036) \approx 0.058$. Interval: 0.747 ± 0.058 , or approximately (0.688, 0.805).

- 52 A poll of 300 randomly selected voters finds that 143 support a candidate. Construct a 95% confidence interval for the true population proportion, using $z^* \approx 1.96$.

- a. (0.450, 0.533)
- b. (0.420, 0.533) ✓
- c. (0.400, 0.553)
- d. (0.420, 0.503)

$\hat{p} = 143/300 = 0.477$. Standard error: $\sqrt{0.477(0.523)/300} \approx 0.029$. Margin of error: $1.96(0.029) \approx 0.057$. Interval: 0.477 ± 0.057 , or approximately (0.420, 0.533).

- 53 A researcher wants a margin of error of at most 0.04 for a 99% confidence interval for a population proportion, using a planning value of $p = 0.30$. Find the minimum sample size needed, using $n = (z^*/m)^2 p(1 - p)$ with $z^* = 2.576$.

- a. 821
- b. 870
- c. 871 ✓
- d. 921

$n = (2.576/0.04)^2(0.30)(0.70) \approx 870.9$. Round up: $n = 871$ (sample size must always be rounded up to guarantee the margin of error requirement is met).

- 54 A researcher wants a margin of error of at most 0.02 for a 90% confidence interval for a population proportion, using a planning value of $p = 0.70$. Find the minimum sample size needed, using $n = (z^*/m)^2 p(1 - p)$ with $z^* = 1.645$.

- a. 1421 ✓
- b. 1420
- c. 1371
- d. 1471

$n = (1.645/0.02)^2(0.70)(0.30) \approx 1420.7$. Round up: $n = 1421$ (sample size must always be rounded up to guarantee the margin of error requirement is met).

- 55 A researcher wants a margin of error of at most 0.03 for a 90% confidence interval for a population proportion, using a planning value of $p = 0.50$. Find the minimum sample size needed, using $n = (z^*/m)^2 p(1 - p)$ with $z^* = 1.645$.

a. 802

b. 752 ✓

c. 751

d. 702

$n = (1.645/0.03)^2(0.50)(0.50) \approx 751.7$. Round up: $n = 752$ (sample size must always be rounded up to guarantee the margin of error requirement is met).

- 56 A researcher wants a margin of error of at most 0.03 for a 95% confidence interval for a population proportion, using a planning value of $p = 0.40$. Find the minimum sample size needed, using $n = (z^*/m)^2 p(1 - p)$ with $z^* = 1.96$.

a. 975

b. 1024

c. 1075

d. 1025 ✓

$n = (1.96/0.03)^2(0.40)(0.60) \approx 1024.4$. Round up: $n = 1025$ (sample size must always be rounded up to guarantee the margin of error requirement is met).

- 57 A researcher wants a margin of error of at most 0.02 for a 90% confidence interval for a population proportion, using a planning value of $p = 0.60$. Find the minimum sample size needed, using $n = (z^*/m)^2 p(1 - p)$ with $z^* = 1.645$.

a. 1574

b. 1624 ✓

c. 1674

d. 1623

$n = (1.645/0.02)^2(0.60)(0.40) \approx 1623.6$. Round up: $n = 1624$ (sample size must always be rounded up to guarantee the margin of error requirement is met).

58 A hypothesis test about a manufacturer's claimed mean product lifetime produces a p -value of 0.047. Using a significance level of $\alpha = 0.01$, state the correct conclusion.

- a. The test is inconclusive and must be repeated
- b. Fail to reject H_0 ; not enough evidence that the true mean lifetime differs from the claim ✓**
- c. Accept H_0 as true
- d. Reject H_0 ; the true mean lifetime differs from the claim

Since the p -value (0.047) is greater than $\alpha = 0.01$, we fail to reject H_0 . Fail to reject H_0 ; not enough evidence that the true mean lifetime differs from the claim.

59 A hypothesis test about a nutritionist's claimed mean calorie count produces a p -value of 0.112. Using a significance level of $\alpha = 0.01$, state the correct conclusion.

- a. The test is inconclusive and must be repeated
- b. Fail to reject H_0 ; not enough evidence that the true mean calorie count differs from the claim ✓**
- c. Reject H_0 ; the true mean calorie count differs from the claim
- d. Accept H_0 as true

Since the p -value (0.112) is greater than $\alpha = 0.01$, we fail to reject H_0 . Fail to reject H_0 ; not enough evidence that the true mean calorie count differs from the claim.

60 A hypothesis test about a claimed mean test score produces a p -value of 0.089. Using a significance level of $\alpha = 0.05$, state the correct conclusion.

- a. Accept H_0 as true
- b. Fail to reject H_0 ; not enough evidence that the true mean test score differs from the claim ✓**
- c. Reject H_0 ; the true mean test score differs from the claim
- d. The test is inconclusive and must be repeated

Since the p -value (0.089) is greater than $\alpha = 0.05$, we fail to reject H_0 . Fail to reject H_0 ; not enough evidence that the true mean test score differs from the claim.

- 61 A hypothesis test about a claimed mean reaction time produces a p -value of 0.034. Using a significance level of $\alpha = 0.05$, state the correct conclusion.
- a. The test is inconclusive and must be repeated
 - b. Accept H_0 as true
 - c. Fail to reject H_0 ; not enough evidence that the true mean reaction time differs from the claim
 - d. **Reject H_0 ; the true mean reaction time differs from the claim ✓**

Since the p -value (0.034) is less than $\alpha = 0.05$, we reject H_0 . Reject H_0 ; the true mean reaction time differs from the claim.

- 62 A hypothesis test about a claimed mean reaction time produces a p -value of 0.089. Using a significance level of $\alpha = 0.01$, state the correct conclusion.
- a. Reject H_0 ; the true mean reaction time differs from the claim
 - b. **Fail to reject H_0 ; not enough evidence that the true mean reaction time differs from the claim ✓**
 - c. Accept H_0 as true
 - d. The test is inconclusive and must be repeated

Since the p -value (0.089) is greater than $\alpha = 0.01$, we fail to reject H_0 . Fail to reject H_0 ; not enough evidence that the true mean reaction time differs from the claim.